THE GUARDIAN – MISINFORMATION DETECTION MODULE

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EVALUATION AND VALIDATION REPORT

This report summarizes the evaluation and validation results of the **Misinformation Detection Module** developed as part of **The Guardian research project**. The module detects misinformation in user-highlighted web content and provides contextual evidence and warnings to users.

1. MODULE ARCHITECTURE OVERVIEW

The system uses a **hybrid architecture** combining a fine-tuned Transformer classifier, risk-aware personalization, and a dual evidence layer.

1.1 END-TO-END PIPELINE

User Text Capture

Users highlight text on any webpage or social platform.
The **Analyze** button in the browser extension sends the selected text to the system for analysis.

User Profile and Risk Context

The extension retrieves user data such as:

- Age group
- Education level

Using these parameters, the system computes:

- **Susceptibility Score (0–100)**
- **Risk Level (Low / Medium / High)**

Offline Classification (Primary Verdict)

The system uses a **DistilBERT sequence classifier** fine-tuned on benchmark misinformation datasets combined with local datasets.

The classifier outputs:

- Predicted label (Misinformation / Factual)
- Confidence score

Evidence Retrieval Layer

Two evidence sources are used:

Local evidence (offline)
TF-IDF similarity search retrieves the most relevant examples from the dataset.

Fact-check evidence (online)
Google Fact Check Tools API returns publisher, rating, title, and URL when available.

All evidence is presented in a single **Evidence** section for user clarity.

Risk-Based Response Generation

Response detail depends on risk level:

Risk Level	Response Type
Low	Short conclusion
Medium	Brief reasoning + verification advice
High	Detailed explanation + strong warnings

UI Rendering

The browser extension popup displays:

- Verdict and confidence score
- Evidence links and example snippets
- Susceptibility score and risk badge

2. DATASET FOR EVALUATION

A **combined dataset** was created by merging benchmark misinformation datasets with locally curated datasets. All data was converted into a unified **JSONL format with binary labels**.

2.1 DATASET SOURCES

The dataset includes samples from the following sources:

- Local CSV dataset
- FakeNewsNet
- GossipCop
- PolitiFact
- LIAR dataset
- CoAID COVID-19 dataset

2.2 DATASET FORMAT

Each JSONL record contains:

- **text** – normalized claim or article text
- **label**
  - 1 = misinformation / fake
  - 0 = factual / real
- **source** – dataset origin

2.3 DATASET SIZE SUMMARY

Combined Dataset (combined.jsonl)

Total samples: **104,518**

Label distribution:

- Factual (0): **58,776**
- Misinformation (1): **45,742**

Dataset Splits

Dataset	Samples
Train	83,613
Validation	10,451
Test	10,454

3. EVALUATION METHODOLOGY

3.1 OFFLINE CLASSIFICATION EVALUATION

The DistilBERT model was evaluated using the held-out test dataset.

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Macro-F1
- Confusion Matrix

These metrics measure both **overall correctness** and **error trade-offs** for misinformation detection.

3.2 EVIDENCE VALIDATION

Evidence generation was tested by:

- Validating returned fact-check citations
- Verifying TF-IDF retrieval of related dataset examples
- Confirming readable evidence display in the extension UI

3.3 END-TO-END LATENCY

Latency measures the time between:

User clicking Analyze → Popup showing result

This includes:

- extension communication time
- classifier inference
- evidence retrieval
- UI rendering

Recommended metrics:

- Mean latency
- P95 latency

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4. FINAL EVALUATION RESULTS

4.1 OVERALL PERFORMANCE

Accuracy: **98.85%**
Macro-F1: **98.85%**

These results exceed the project requirement of **≥ 85% accuracy**.

4.2 CONFUSION MATRIX

```
[[3424, 29],
 [  50, 3358]]
```

Interpretation:

- True Negatives: 3424
- False Positives: 29
- False Negatives: 50
- True Positives: 3358

4.3 PRECISION, RECALL, AND F1

Precision: **0.991**
Recall: **0.985**
F1-Score: **0.988**

All metrics exceed the required performance threshold.

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5. ERROR ANALYSIS

Observed errors:

- False positives: **29**
- False negatives: **50**

Common causes include:

- Limited context in highlighted text
- Ambiguous claims
- New misinformation patterns
- Sarcasm or rhetorical statements
- Informal social media language

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6. KEY IMPROVEMENTS IMPLEMENTED

Major improvements include:

- Offline DistilBERT misinformation detection
- Multi-dataset training (LIAR, FakeNewsNet, CoAID, Local)
- Risk-based personalization using susceptibility scores
- Dual evidence layer (offline examples + fact-check API)
- Improved browser extension user experience

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7. CONCLUSION

The evaluation results demonstrate that the module achieves **high-accuracy misinformation detection** with strong **precision, recall, and F1 performance**.

The dual evidence mechanism enables:

- Offline evidence using dataset examples
- Verified fact-check sources when available

Therefore, the module provides a **reliable foundation for real-time misinformation warnings in browsing environments**.